JIAYU (RYAN) ZUO

Junior Undergraduate
College of Engineering · Automation

HomePage
Education Email
+86 18079101014
Kuni05
Shenzhen, China
Jiayu Zuo

SELF-INTRODUCTION

Hi! I'm a junior undergraduate student majoring in Automation in the College of Engineering at Southern University of Science and Technology (SUSTech). Strong interests in neuroscience and BCI, with goals to explore the mysteries of the brain and develop innovative brain-computer interface technologies.

Currently conducting research at NCC Lab under the supervision of Associate Professor Quanying Liu. For my latest information, please see my personal homepage.

SKILLS

Programming: Python, C, Java, MATLAB, LaTeX

Frameworks: PyTorch, NumPy, MNE

Tools: Git, Docker, Jupyter

Languages: Chinese (Native)
English (Fluent, CET-4)

PUBLICATIONS & UNDER REVIEW

**Equal Contribution*

BHA *(title abbreviated to avoid blind review issues)*
D. Li*, P. Jiang*, **J. Zuo***, C. Wei, M. Wang, L. Wang, X. Jia, & Q. Liu — *Under Review (ICLR 2027)*

BrainGrasper *(title abbreviated to avoid blind review issues)*
J. Zuo*, P. Chen*, J. Li, X. Li, & Q. Liu — *Under Review (ICRA 2027)*

BrainXPO *(title abbreviated to avoid blind review issues)*
J. Zuo*, D. Li*, H. Zhang, S. Chen, W. Yang, Y. Liu, & C. Wei — *Under Review (AAAI 2027)*

BrainTaskonomy *(title abbreviated to avoid blind review issues)*
J. Xia, W. Ye, J. Zhang, **J. Zuo**, M. Wang, & Q. Liu — *Under Review (AAAI 2027)*

FlatClip: Reusing Image Foundation Models for fMRI Representation Learning via Cortical Flatmaps
M. Wang*, W. Ye*, Z. Ning, **J. Zuo**, J. Xia, H. Wen, & Q. Liu — *NeurIPS 2026, Poster*

RESEARCH PROJECTS

Leading Projects

07/2026 – Present **Human Preference-Aligned Generative Model Post-Training** — **Omni-Intelligence**
- Conducting research on human preference-aligned post-training for generative models, focusing on alignment and model behavior optimization.

`Post-Training` / `Human Preference Alignment` / `Generative Models` / `Reinforcement Learning`

01/2026 – Present **EEG-Based Visual Decoding in Real-World Natural Scenes** — **NCC Lab, SUSTech**
- Designed a decoding framework targeting few-trial EEG responses to complex natural scene stimuli, reducing reliance on trial-averaging that limits real-world applicability.
- Proposed a subject-agnostic alignment strategy to achieve robust cross-subject generalization without per-subject fine-tuning.

`Few-Trial Decoding` / `Cross-Subject Generalization` / `Natural Scene` / `EEG-to-Image`

09/2025 – 07/2026 **EEG-to-Text Decoding via Foundation Model Alignment** — **NCC Lab, SUSTech**
- Developed a brain-to-text decoding pipeline aligning EEG representations with large language model embedding spaces via contrastive and generative objectives.
- Extended decoding to multilingual settings, bridging neural EEG signals and natural language text across languages.
- Funded by the Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation ("Climbing Program" Special Funds).

`EEG-to-Text` / `Contrastive Learning`

06/2025 – 09/2025 **Multi-View 3D Reconstruction System for EEG Electrode Localization** — **NCC Lab, SUSTech**
- Developed a precise electrode positioning system using multi-view 3D reconstruction techniques
- Implemented source localization algorithms and VGGT for visual reconstruction
- Applied YOLO-based object detection for automated electrode identification
- Filed one patent application (under review)

`3D Gaussian Splatting` / `YOLO` / `Source Localization`

Participant Projects

03/2026 – 05/2026 **Adolescent Brain Development Assessment via Large-Scale fMRI Foundation Model** **NCC Lab, SUSTech**

- Building a foundation model pre-trained on a large-scale private fMRI dataset of Chinese school-age children and adolescents to capture developmental brain patterns.
- Designing downstream evaluation tasks for cognitive and neurodevelopmental assessment, including age prediction and developmental trajectory modeling.

`fMRI Foundation Model / Brain Development`

EDUCATION

08/2024 - 06/2028 **Bachelor of Engineering (BEng)** **Southern University of Science and Technology (SUSTech)**

Major: Automation

Completed: Calculus, Linear Algebra, College Physics, Fundamentals of Electronic Circuits

In Progress: Data Structures and Algorithms, Probability & Statistics, Signals and Systems

HONORS & AWARDS

05/2026 **College Students' Innovative Entrepreneurial Training Plan Program** **SUSTech**

- Project: *Opti-Rhythm: A Digital Sleep Modulation System Based on a Physiological-Topology Brain-Body Coupling Foundation Model*.
- Selected with a grant of **CNY 20,000**.

12/2025 **Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation** ("Climbing Program" Special Funds) **SUSTech**

- Project: *Foundation Model-Aligned Chinese Semantic Decoding from Brain Signals*.
- Selected as a top-tier undergraduate project (Top 30%) with a grant of **CN¥20,000**.

11/2025 **Award of Excellence (Poster Presentation)** **Department of Biomedical Engineering, SUSTech**

- Title: *MindPilot: EEG-guided Brain Modulation using Closed-loop Visual Stimulation Optimization*.
- Awarded at the 6th SUSTech BME Research Day.

COMPETITIONS

07/2026 **The 2026 NeuroGolf Championship, Bronze Medal (Rank 193/3061)** **Kaggle**

- **Solo Participant**. Task: *Design the smallest neural networks to solve ARC-AGI image transformations*.

05/2026 **Finalist (About Top 1%), 2026 Mathematical Contest in Modeling (MCM)** **COMAP**

- **Team Leader & Advisor**. Topic: *"Artemis's Choice: Chariot or Elevator?"*.

12/2025 **Winning Prize, 2025(6th) "Greater Bay Area Cup" Financial Mathematical Modeling Contest** **GDSTA**

- **Team Leader**. Topic: *"Comprehensive Evaluation and Development Analysis of Stablecoins"*.
- Hosted by Guangdong Society for Industrial and Applied Mathematics & Guangdong-Hong Kong-Macau Center for Applied Mathematics(Sun Yat-sen University).

05/2025 **Honorable Mention (H Award), 2025 Mathematical Contest in Modeling (MCM)** **COMAP**

- **Solo Participant**. Topic: *"Cybersecurity Analysis Based on the Global Cybersecurity Index (GCI)"*.

2023 **National Second Prize, China High School Mathematical Contest in Modeling** **Beijing Normal University**

- **Solo Participant**. Topic: *"Game Theoretic Analysis of Medical Resource Runs During the COVID-19 Pandemic"*.

RESEARCH EXPERIENCE

Omni-Intelligence *Shenzhen*

06/2026 – Present *Supervisors: Dr. Chen Wei & Mr. Dongyang Li*

Research Intern working on human preference-aligned post-training for generative models.

Neural Computing and Control Lab (NCC Lab) *Department of Biomedical Engineering, SUSTech*

09/2024 – Present *Supervisor: Associate Professor Quanying Liu*

Undergraduate Researcher focusing on BCI, CHI, and NeuroAI.